Solutions for Assignment-2 (Semiconductor Diode)

1. Material Differences: n-type semiconductors are doped with pentavalent atoms (like Phosphorus or Arsenic) which introduce an excess of free electrons (negative charge carriers). Conversely, p-type semiconductors are doped with trivalent atoms (like Boron or Gallium), creating an excess of 'holes' (the absence of an electron, which behaves as a positive charge carrier). In n-type, electrons are the majority carriers and holes are minority carriers; in p-type, holes are the majority carriers and electrons are minority carriers.

2. p-n Junction Bias Conditions: Under forward-bias, the positive terminal is connected to the p-side and the negative to the n-side. This narrows the depletion region and reduces the internal electric field, allowing majority carriers (electrons and holes) to cross the junction in large numbers, resulting in a large, exponentially increasing current flow.

Under reverse-bias, the terminal connections are flipped (negative to p-side, positive to n-side). This widens the depletion region, strengthening the potential barrier and effectively blocking the flow of majority carriers. The resulting current is extremely small and is called the saturation current, I_s, composed primarily of minority carriers.

3. Meaning of 'Ideal': The term 'ideal' is used to describe a theoretical model of a component with perfect, simplified characteristics, ignoring non-ideal effects like resistance, heating, or leakage for the sake of simplified analysis. For an ideal diode, for instance, it is assumed to have zero voltage drop and zero resistance when forward-biased, and infinite resistance with zero current when reverse-biased, effectively acting as a perfect closed or open switch.

4. (a) Thermal Voltage:

We determine the thermal voltage (V_T) for the given temperature. First, we convert the temperature from Celsius to Kelvin using $T_K = T_C + 273$.

$$T_K = 25°\text{C} + 273 = 298\text{K}.$$

Next, we apply the formula $V_T = \frac{kT}{q}$, where k is Boltzmann's constant, T is temperature in Kelvin, and q is the electron charge.

$$k = 1.38 \times 10^{-23}\ \text{J/K}$$
$$q \approx 1.6 \times 10^{-19}\ \text{C}$$
$$V_T = \frac{(1.38 \times 10^{-23}\ \text{J/K}) \cdot 298\text{K}}{1.6 \times 10^{-19}\ \text{C}}$$
$$V_T = \frac{4.1124 \times 10^{-21}}{1.6 \times 10^{-19}}\ \text{V}$$
$$V_T \approx 0.0257\ \text{V or } 25.7\text{mV}$$

Final Answer (a): The thermal voltage is approximately $V_T \approx 0.0257$ V or 25.7mV.

(b) Diode Current:

Now, we calculate the diode current (I_D) using the diode equation: $I_D = I_s\left(e^{\frac{V_D}{nV_T}} - 1\right)$

$$I_s = 40\text{nA} = 40 \times 10^{-9}\ \text{A}$$
$$n = 2$$
$$V_D = 0.5\ \text{V}$$
$$V_T = 0.0257\ \text{V (from part a)}$$

First, let's calculate the exponent:

$$\text{Exponent} = \frac{V_D}{n \cdot V_T} = \frac{0.5}{2 \cdot 0.0257} \approx \frac{0.5}{0.0514} \approx 9.73$$

Now, substitute all values back into the equation:

$$I_D = (40 \times 10^{-9}) \cdot (e^{9.73} - 1)\ \text{A}$$
$$I_D \approx (40 \times 10^{-9}) \cdot (16815 - 1)\ \text{A}$$
$$I_D \approx (40 \times 10^{-9}) \cdot 16814\ \text{A}$$
$$I_D \approx 0.000673\ \text{A or } 0.673\text{mA}$$

Final Answer (b): The diode current is approximately $I_D \approx 0.673$mA.